

Machine Learning in Finance & Insurance

Fall 2026, ETH Zurich

1. Basic Notions of Statistical Learning

- 1.1 The Prediction Problem
- 1.2 Regression Functions and Irreducible Error
- 1.3 Empirical Loss
- 1.4 Hypothesis Classes
- 1.5 Bias-Variance Tradeoff
- 1.6 R^2
- 1.7 Decomposition of the Generalization Error
- 1.8 Training Error and Test Error
- 1.9 Regularization

1.1 The Prediction Problem

Consider ...

- a *probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, that is,
 - Ω is a non-empty set (*sample space*)
 - $\mathcal{F}$ is a *σ -algebra* of subsets of Ω
 - $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a *probability measure*

- a *feature space* $\mathcal{X} \subseteq \mathbb{R}^d$ with *Borel σ -algebra* $\mathcal{B}_{\mathcal{X}}$

e.g. $\mathcal{X} = \mathbb{R}^m \times \{1, 2, \dots, n\} \times \mathbb{N}$

- a closed or open *target space* $\mathcal{Y} \subseteq \mathbb{R}^k$ with *Borel σ -algebra* $\mathcal{B}_{\mathcal{Y}}$

e.g. $\mathcal{Y} = \mathbb{R}, (0, 1)$ or $\{0, 1\}$

- random elements $X : \Omega \rightarrow \mathcal{X}$ (*input variables/independent variables/features/covariates*)

$Y : \Omega \rightarrow \mathcal{Y}$ (*output variable/dependent variable/response variable*)

Let ...

- ρ_X, ρ_Y, ρ be the *distributions* of $X, Y, (X, Y)$ on $\mathcal{X}, \mathcal{Y}, \mathcal{X} \times \mathcal{Y}$, respectively
- $\rho_{Y|X}$ a *regular conditional distribution* of Y given X ; that is,
 - for all $x \in \mathcal{X}$, $\rho_{Y|X}(\cdot | x)$ is a probability measure on $\mathcal{Y}$
 - for all $B \in \mathcal{B}_{\mathcal{Y}}$, $x \mapsto \rho_{Y|X}(B | x)$ is $\mathcal{B}_{\mathcal{X}}$ -measurable
 - for all $B \in \mathcal{B}_{\mathcal{Y}}$, $\mathbb{E}[1_B(Y) | X] = \rho_{Y|X}(B | X)$ $\mathbb{P}$ -almost surely

it follows (from a monotone class argument) that for all ρ -integrable $h: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$,

$$\int_{\mathcal{X} \times \mathcal{Y}} h(x, y) \rho(dx, dy) = \int_{\mathcal{X}} \int_{\mathcal{Y}} h(x, y) \rho_{Y|X}(dy | x) \rho_X(dx) \quad (\text{disintegration of } \rho)$$

- *Prediction problem*: find a measurable function $f: \mathcal{X} \rightarrow \mathcal{Y}$ such that $f(X) \approx Y$, e.g.
 - if $\mathcal{Y} \subseteq \mathbb{R}$, try to minimize $\mathbb{E}[(f(X) - Y)^2]$ or $\mathbb{E}|f(X) - Y|$
 - in general, try to minimize the *expected loss (or risk)*

$$\mathbb{E} \ell(f(X), Y) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \rho(dx, dy)$$

for a measurable *loss function* $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$

Trivial example: Y independent of X

- Assume X and Y are independent, $\mathcal{Y} = \mathbb{R}$, $\mathbb{E}[Y^2] < \infty$ and $\ell(w, y) = (w - y)^2$
- Then, for any measurable function $f : \mathcal{X} \rightarrow \mathcal{Y}$, the **Mean Squared Error (MSE)** is

$$\mathbb{E} \left[(f(X) - Y)^2 \right] = (\mathbb{E}[f(X) - Y])^2 + \text{Var}(f(X) - Y) = (\mathbb{E}[f(X) - Y])^2 + \text{Var}(f(X)) + \text{Var}(Y)$$

$\uparrow$ $\uparrow$ $\uparrow$
bias² variance irreducible error

- **Theoretically best predictor:** $\bar{f}(X) = \mathbb{E} Y$ (is known if the distribution of Y is known)

$$\text{Bias}(\bar{f}(X)) = \mathbb{E}[\bar{f}(X) - Y] = \text{Var}(\bar{f}(X)) = 0, \quad \text{MSE} = \mathbb{E}[(Y - \mathbb{E} Y)^2] = \text{Var}(Y)$$

- If $\mathbb{E} Y$ is unknown, it can be estimated from realizations of Y
- Let $Y_i, i = 1, \dots, m$, be independent copies of Y (introduces additional randomness)
- In ML, it is common to solve the **empirical loss/risk minimization problem**

$$\min_{g : \mathbb{R}^m \rightarrow \mathbb{R}} \frac{1}{m} \sum_{i=1}^m (g(Y_{1:m}) - Y_i)^2, \quad \text{where } Y_{1:m} = (Y_1, \dots, Y_m)$$

- $$\frac{1}{m} \sum_{i=1}^m (g(Y_{1:m}) - Y_i)^2 = g^2(Y_{1:m}) - 2g(Y_{1:m}) \frac{1}{m} \sum_{i=1}^m Y_i + \frac{1}{m} \sum_{i=1}^m Y_i^2$$

$$= \left(g(Y_{1:m}) - \frac{1}{m} \sum_{i=1}^m Y_i \right)^2 + \frac{1}{m} \sum_{i=1}^m Y_i^2 - \left(\frac{1}{m} \sum_{i=1}^m Y_i \right)^2$$
- $\rightsquigarrow$ the **empirical loss/risk minimizer** is $\bar{g}(Y_{1:m}) = \frac{1}{m} \sum_{i=1}^m Y_i$ (**empirical mean**)
- for any measurable function $g : \mathbb{R}^m \rightarrow \mathbb{R}$, the **MSE** is

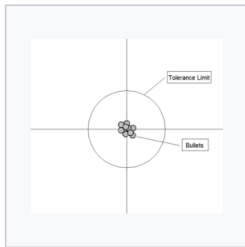
$$\begin{aligned} \mathbb{E} \left[(g(Y_{1:m}) - Y)^2 \right] &= (\mathbb{E}[g(Y_{1:m}) - Y])^2 + \text{Var}(g(Y_{1:m}) - Y) \\ &= (\underbrace{\mathbb{E}[g(Y_{1:m}) - Y]}_{\text{bias}^2})^2 + \underbrace{\text{Var}(g(Y_{1:m}))}_{\text{variance}} + \underbrace{\text{Var}(Y)}_{\text{irreducible error}} \end{aligned}$$

- there is a **Bias-Variance Tradeoff**; the best balance depends on the (unknown) distribution of Y !
- e.g. $\text{Bias}(\bar{g}(Y_{1:m})) = \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m Y_i - Y \right] = 0$, $\text{Var}(\bar{g}(Y_{1:m})) = \frac{1}{m^2} \sum_{i=1}^m \text{Var}(Y_i) = \frac{1}{m} \text{Var}(Y)$

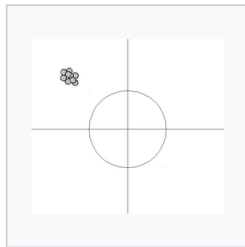
the **empirical mean** has no bias, it has a small MSE if $\text{Var}(Y)/m$ is small

otherwise, e.g. the **empirical median** might be better (has a bias but lower variance)

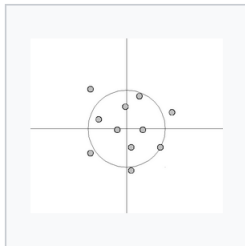
or **median of means of subgroups**



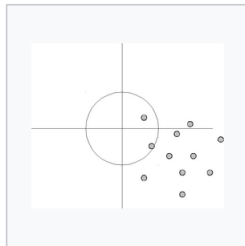
Low bias, low variance



High bias, low variance



Low bias, high variance



High bias, high variance

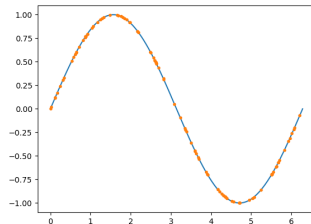
Examples with dependence between X and Y

- Deterministic dependence: $Y = g(X)$,

e.g. $X \sim \text{Unif}(0, 2\pi)$, $Y = \sin(X)$

$(X_i, Y_i), i = 1, \dots, 100$,

indep. copies of (X, Y)



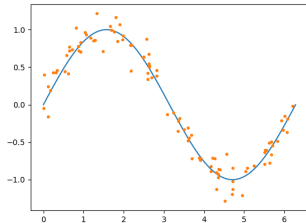
- Homoscedastic additive noise: $Y = g(X) + \varepsilon$ for ε independent of X with $\mathbb{E}[\varepsilon] = 0$,

e.g. $X \sim \text{Unif}[0, 2\pi]$, $Y = \sin(X) + \varepsilon$,

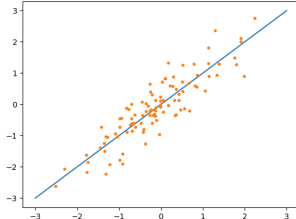
$\varepsilon \sim \mathcal{N}(0, \sigma^2)$,

or $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_d X_d + \varepsilon$,

$X \sim \mathcal{N}_d(0, I_d)$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$,



(X_i, Y_i) ,
 $i = 1, \dots, 100$,
indep. copies
of (X, Y)

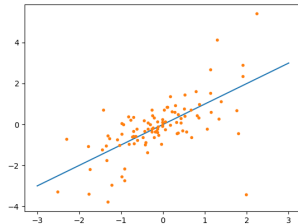


(X_i, Y_i) ,
 $i = 1, \dots, 100$,
indep. copies
of (X, Y)

- **Heteroscedastic additive noise:** $Y = g(X) + h(X)\varepsilon$ for ε independent of X with $\mathbb{E}[\varepsilon] = 0$,

e.g. $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_d X_d + (1 + \|X\|_2^2)\varepsilon$
 for $X \sim N_d(0, I_d)$ and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

$(X_i, Y_i), i = 1, \dots, 100$, indep. copies of (X, Y)



- **Non-additive noise:** $Y = g(X, \varepsilon)$ for ε independent of X ,

e.g. $Y | X \sim \text{Ber}(p(X))$ for a function $p : \mathcal{X} \rightarrow [0, 1]$ (**Bernoulli with success probability $p(X)$**),

that is,
$$\begin{cases} \mathbb{P}[Y = 1 | X] = p(X) \\ \mathbb{P}[Y = 0 | X] = 1 - p(X) \end{cases} \quad \text{(used in binary classification; e.g. fraud detection)}$$

- **Model-free:** the form of the underlying model is not known, only iid observations $(X_i, Y_i)_{i=1}^m$ are available

1.2 Regression Functions and Irreducible Error

Lemma (Theoretically best predictor with respect to a given loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$)

Let $\bar{f} : \mathcal{X} \rightarrow \mathcal{Y}$ be a measurable function such that for ρ_X -almost all $x \in \mathcal{X}$,

$\bar{f}(x)$ is a minimizer of the mapping $w \mapsto \int_{\mathcal{Y}} \ell(w, y) \rho_{Y|X}(dy | x) \in [0, \infty]$.

Then $\mathbb{E} \ell(\bar{f}(X), Y) \leq \mathbb{E} \ell(f(X), Y)$ for any measurable function $f : \mathcal{X} \rightarrow \mathcal{Y}$

($\mathbb{E} \ell(\bar{f}(X), Y)$ might be ∞)

Proof
$$\begin{aligned} \mathbb{E} \ell(\bar{f}(X), Y) &= \int_{\mathcal{X} \times \mathcal{Y}} \ell(\bar{f}(x), y) \rho(dx, dy) = \int_{\mathcal{X}} \int_{\mathcal{Y}} \ell(\bar{f}(x), y) \rho_{Y|X}(dy | x) \rho_X(dx) \\ &\leq \int_{\mathcal{X}} \int_{\mathcal{Y}} \ell(f(x), y) \rho_{Y|X}(dy | x) \rho_X(dx) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \rho(dx, dy) = \mathbb{E} \ell(f(X), Y) \end{aligned}$$

□

From now on, we suppose the assumptions of the lemma hold, and we call ...

- $\bar{f}$ *regression function*
- $\mathbb{E} \ell(\bar{f}(X), Y)$ *irreducible error*

e.g. minimal mean squared error (**MSE**) if $\mathcal{Y} \subseteq \mathbb{R}$ and $\ell(w, y) = (w - y)^2$

or minimal mean absolute error (**MAE**) if $\mathcal{Y} \subseteq \mathbb{R}$ and $\ell(w, y) = |w - y|$

Example 1: Square Loss: $\mathcal{Y} \subseteq \mathbb{R}$ and $\ell(w, y) = (w - y)^2$

- Consider a random variable Z with $\mathbb{E}[Z^2] < \infty$
- Then $\mathbb{E}|Z| < \infty$, and for all $w \in \mathbb{R}$,
$$\mathbb{E}[(w - Z)^2] = w^2 - 2w\mathbb{E}[Z] + \mathbb{E}[Z^2]$$
$$= w^2 - 2w\mathbb{E}[Z] + \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 + (\mathbb{E}[Z])^2 = (w - \mathbb{E}[Z])^2 + \text{Var}(Z)$$
- So the mapping $w \mapsto \mathbb{E}(w - Z)^2$ is strictly convex with ...
 - unique minimizer $w = \mathbb{E} Z$ and
 - minimal value $\text{Var}(Z) = \mathbb{E}[(Z - \mathbb{E} Z)^2] = \mathbb{E} Z^2 - (\mathbb{E} Z)^2$

- Now, assume $\infty > \mathbb{E}[Y^2] = \int_{\mathcal{X} \times \mathcal{Y}} y^2 \rho(dx, dy) = \int_{\mathcal{X}} \int_{\mathcal{Y}} y^2 \rho_{Y|X}(dy, x) \rho_X(dx)$

Then $\int_{\mathcal{Y}} y^2 \rho_{Y|X}(dy, x) < \infty$, and therefore, $\int_{\mathcal{Y}} |y| \rho_{Y|X}(dy, x) < \infty$ for ρ_X -almost all $x \in \mathcal{X}$

- For all $x \in \mathcal{X}$, the unique minimizer $\bar{f}(x)$ of $w \mapsto \int_{\mathcal{Y}} (w - y)^2 \rho_{Y|X}(dy, x)$ is
$$\bar{f}(x) = \int_{\mathcal{Y}} y \rho_{Y|X}(dy, x) = \mathbb{E}[Y | X = x]$$

- Therefore, $\bar{f}(X) = \mathbb{E}[Y | X]$ and
$$\mathbb{E} \ell(\bar{f}(X), Y) = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2]$$
$$= \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X]] = \mathbb{E} \text{Var}(Y | X) = \text{Var}(Y) - \text{Var}(\mathbb{E}[Y | X])$$

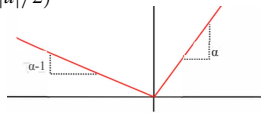
(Least Squares Regression)

Example 2: Pinball Loss: $\mathcal{Y} \subseteq \mathbb{R}$ and $\ell(w, y) = \ell_\alpha(y - w)$

- Let $\alpha \in (0, 1)$ and $\ell(w, y) = \ell_\alpha(y - w)$ for the **pinball loss function** $\ell_\alpha(u) = (\alpha - 1_{\{u < 0\}}) u$, $u \in \mathbb{R}$

(for $\alpha = 1/2$, $\ell_\alpha(u) = |u|/2$)

- Note that $\partial_+ \ell_\alpha(u) = \alpha - 1_{\{u < 0\}}$ and $\partial_- \ell_\alpha(u) = \alpha - 1_{\{u \leq 0\}}$
- Consider a random variable Z with $\mathbb{E}|Z| < \infty$
- Then $\mathbb{E} \ell_\alpha(Z - w)$ is convex in w and



$$\partial_+ \mathbb{E} \ell_\alpha(Z - w) = -\mathbb{E} \partial_- \ell_\alpha(Z - w) = -\mathbb{E} [\alpha - 1_{\{Z - w \leq 0\}}] = -\alpha + \mathbb{P}[Z \leq w]$$

$$\partial_- \mathbb{E} \ell_\alpha(Z - w) = -\mathbb{E} \partial_+ \ell_\alpha(Z - w) = -\mathbb{E} [\alpha - 1_{\{Z - w < 0\}}] = -\alpha + \mathbb{P}[Z < w]$$

- So, w is a minimizer of $\mathbb{E} \ell_\alpha(Z - w) \Leftrightarrow \mathbb{P}[Z < w] \leq \alpha$ and $\mathbb{P}[Z \leq w] \geq \alpha \Leftrightarrow w$ is an α -quantile of Z

- Now, assume $\infty > \mathbb{E}|Y| = \int_{\mathcal{X} \times \mathcal{Y}} |y| \rho(dx, dy) = \int_{\mathcal{X}} \int_{\mathcal{Y}} |y| \rho_{Y|X}(dy, x) \rho_X(dx)$

Then $\int_{\mathcal{Y}} |y| \rho_{Y|X}(dy, x) < \infty$ for ρ_X -almost all $x \in \mathcal{X}$

- Let $q_\alpha(x)$, for $x \in \mathcal{X}$, be an α -quantile of the conditional distribution $\rho_{Y|X}(\cdot, x)$ of Y given $X = x$
- Then, $\bar{f} = q_\alpha$ and $\mathbb{E} \ell(\bar{f}(X), Y) = \mathbb{E} \ell_\alpha(Y - q_\alpha(X))$ (**Quantile Regression**)
- For $\alpha = 1/2$, $q_{1/2}(X)$ = conditional median and $\mathbb{E} \ell_\alpha(Y - q_\alpha(X)) = \frac{1}{2} \mathbb{E} |Y - q_{1/2}(X)|$ (**minimal MAE**)

1.3 Empirical Loss (or Empirical Risk)

- Let $(X_i, Y_i), i = 1, \dots, m$, be independent copies of (X, Y) (*training data*)

- The *empirical loss/risk* of a measurable function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is $\frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) \approx \mathbb{E} \ell(f(X), Y)$

- If $\mathbb{E} \ell^2(f(X), Y) < \infty$, one has

$$\begin{aligned} \mathbb{E} \left[\left(\frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) - \mathbb{E} \ell(f(X), Y) \right)^2 \right] &= \text{Var} \left(\frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) \right) \\ &= \frac{1}{m^2} \sum_{i=1}^m \text{Var}(\ell(f(X_i), Y_i)) = \frac{1}{m} \text{Var}(\ell(f(X), Y)) \xrightarrow{(m \rightarrow \infty)} 0 \quad (\text{Weak Law of Large Numbers}) \end{aligned}$$

1.4 Hypothesis Classes

- A **hypothesis class** is a family $\mathcal{H}$ of measurable functions $f : \mathcal{X} \rightarrow \mathcal{Y}$, e.g.
 - all affine functions: $f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_d x_d$ (if $\mathcal{X} \subseteq \mathbb{R}^d$)
 - 2nd order polynomials: $f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_{1,1} x_1^2 + \dots + \beta_{d-1,d} x_{d-1} x_d$ (if $\mathcal{X} \subseteq \mathbb{R}^d$)
 - splines
 - trees
 - support vector machines
 - neural networks
- Ideally, one would like to minimize $\mathbb{E} \ell(f(X), Y)$ over all measurable functions $f : \mathcal{X} \rightarrow \mathcal{Y}$
- In practice, one tries to find a **numerical solution** $\hat{f}_m$ of the
empirical loss/risk minimization problem
$$\min_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i)$$
 - If the optimization algorithm is deterministic, $\hat{f}_m$ is of the form $\hat{f}_m(\cdot) = \hat{\varphi}(\cdot, (X_i, Y_i)_{i=1}^m)$ for a measurable function $\hat{\varphi} : \mathcal{X} \times (\mathcal{X} \times \mathcal{Y})^m \rightarrow \mathcal{Y}$ (**data-dependent**)
 - If the optimization algorithm is stochastic (**e.g. random initialization**), $\hat{f}_m$ is of the form $\hat{f}_m(\cdot) = \hat{\varphi}(\cdot, (X_i, Y_i)_{i=1}^m, V)$ for a measurable function $\hat{\varphi}$ and a random element V independent of (X, Y) (**data-dependent and noisy**)

Expected Loss Decomposition for the Square Loss Function $\ell(w, y) = (w - y)^2$

- Conditional expected loss given that $X = x$:* $\mathbb{E} \left[\left(\hat{f}_m(X) - Y \right)^2 \mid X = x \right]$

$$= \mathbb{E} \int_{\mathcal{Y}} \left(\hat{f}_m(x) - y \right)^2 \rho_{Y|X}(dy \mid x) = \int_{\mathcal{Y}} \left(\mathbb{E} \hat{f}_m^2(x) - 2\mathbb{E} \hat{f}_m(x)y + y^2 \right) \rho_{Y|X}(dy \mid x)$$

denote $\hat{f}_m^{\text{avg}}(x) = \mathbb{E} \hat{f}_m(x)$ and $\bar{f}(x) = \int_{\mathcal{Y}} y \rho_{Y|X}(dy \mid x)$

$$= \underbrace{\left(\hat{f}_m^{\text{avg}}(x) - \bar{f}(x) \right)^2}_{\substack{\uparrow \\ \text{(conditional) bias}^2}} + \underbrace{\mathbb{E} \left[\left(\hat{f}_m(x) - \hat{f}_m^{\text{avg}}(x) \right)^2 \right]}_{\substack{\uparrow \\ \text{(conditional) variance}}} + \underbrace{\int_{\mathcal{Y}} \left(\bar{f}(x) - y \right)^2 \rho_{Y|X}(dy \mid x)}_{\substack{\uparrow \\ \text{(conditional) irreducible error}}}$$

- Expected loss:* $\mathbb{E} \left[\left(\hat{f}_m(X) - Y \right)^2 \right] = \int_{\mathcal{X}} \mathbb{E} \left[\left(\hat{f}_m(X) - Y \right)^2 \mid X = x \right] \rho_X(dx)$

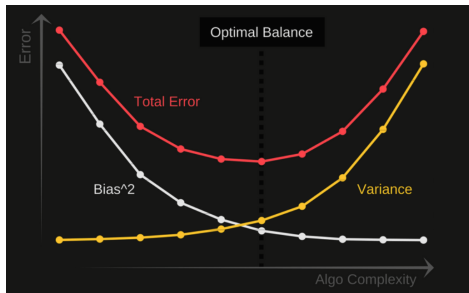
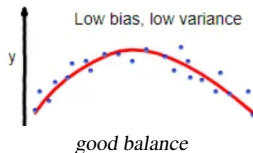
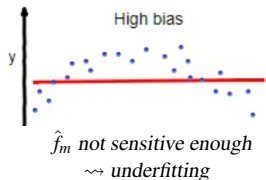
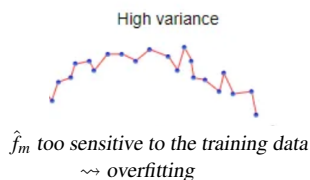
$$= \int_{\mathcal{X}} \left\{ \left(\hat{f}_m^{\text{avg}}(x) - \bar{f}(x) \right)^2 + \mathbb{E} \left[\left(\hat{f}_m(x) - \hat{f}_m^{\text{avg}}(x) \right)^2 \right] + \int_{\mathcal{Y}} \left(\bar{f}(x) - y \right)^2 \rho_{Y|X}(dy \mid x) \right\} \rho_X(dx)$$

$$= \underbrace{\mathbb{E} \left[\left(\hat{f}_m^{\text{avg}}(X) - \bar{f}(X) \right)^2 \right]}_{\substack{\uparrow \\ \text{(average) bias}^2}} + \underbrace{\mathbb{E} \left[\left(\hat{f}_m(X) - \hat{f}_m^{\text{avg}}(X) \right)^2 \right]}_{\substack{\uparrow \\ \text{(average) variance}}} + \underbrace{\mathbb{E} \left[\left(\bar{f}(X) - Y \right)^2 \right]}_{\substack{\uparrow \\ \text{irreducible error}}}$$

1.5 Bias-Variance Tradeoff

(for a given size m of the training set $(X_i, Y_i), i = 1, \dots, m$)

- **Bias** = distance of the **average prediction function** $\hat{f}_m^{\text{avg}} = \mathbb{E} \hat{\varphi}(\cdot, (X_i, Y_i)_{i=1}^m, V)$ from the **true regression function** $\bar{f}$
- **Variance** = variation of the **data-dependent (noisy) prediction function** $\hat{f}_m = \hat{\varphi}(\cdot, (X_i, Y_i)_{i=1}^m, V)$



1.6 R^2

- Consider the square loss function $\ell(w, y) = (w - y)^2$
- **Training data:** (X_i, Y_i) , $i = 1, \dots, m$: independent copies of (X, Y)
Test data: (X_i, Y_i) , $i = m + 1, \dots, m + n$: independent copies of (X, Y) independent of (X_i, Y_i) , $i = 1, \dots, m$
- Let $\hat{f}_m : \mathcal{X} \rightarrow \mathcal{Y}$ be a prediction function trained on the training data
- $R^2 = 1 - \frac{SSR}{SST}$, where
 - $SSR = \sum_{i=1}^m (Y_i - \hat{f}_m(X_i))^2$ (*Sum of Squared Residuals*)
 - $SST = \sum_{i=1}^m (Y_i - \bar{Y}_{\text{train}})^2$ for $\bar{Y}_{\text{train}} = \frac{1}{m} \sum_{i=1}^m Y_i$ (*Total Sum of Squares*)

R^2 measures goodness of in-sample fit (the higher the better)

- Out-of-sample R^2 : $R_{os}^2 = 1 - \frac{SSR_{os}}{SST_{os}}$, where
 - $SSR_{os} = \sum_{i=m+1}^{m+n} (Y_i - \hat{f}_m(X_i))^2$
 - $SST_{os} = \sum_{i=m+1}^{m+n} (Y_i - \bar{Y}_{\text{train}})^2$

(out-of-sample versions of SSR and SST)

R_{os}^2 measures prediction power (the higher the better)

Approximation Error and Sampling Error

- **Approximation error:** $\inf_{f \in \mathcal{H}} \mathbb{E} \ell(f(X), Y) - \mathbb{E} \ell(\bar{f}(X), Y) \geq 0$
 - occurs if $\mathcal{H}$ is not flexible enough to approximate $\bar{f}$ well
 - is decreasing if $\mathcal{H}$ is increasing
 - = 0 if $\bar{f} \in \mathcal{H}$
- If the **empirical loss minimization problem** $\min_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i)$ has a solution, we call it $\hat{f}_m^{\mathcal{H}} \in \mathcal{H}$
- If the minimum is not attained, let $\hat{f}_m^{\mathcal{H}} \in \mathcal{H}$ be such that
$$\frac{1}{m} \sum_{i=1}^m \ell(\hat{f}_m^{\mathcal{H}}(X_i), Y_i) \leq \inf_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) + 10^{-100}$$
- $\hat{f}_m^{\mathcal{H}}$ is **random** since it depends on $(X_1, Y_1), \dots, (X_m, Y_m)$
- **Sampling error:** $\mathbb{E} \left[\ell \left(\hat{f}_m^{\mathcal{H}}(X), Y \right) \mid \hat{f}_m^{\mathcal{H}} \right] - \inf_{f \in \mathcal{H}} \mathbb{E} [\ell(f(X), Y)]$
$$= \int_{\mathcal{X} \times \mathcal{Y}} \ell \left(\hat{f}_m^{\mathcal{H}}(x), y \right) \rho(dx, dy) - \inf_{f \in \mathcal{H}} \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \rho(dx, dy) \geq 0$$
 - results from minimizing the **empirical loss** instead of the **expected loss**
 - tends to decrease if the sample size m is increasing

Optimization Error

- Let $\hat{f}_m$ be a *numerical solution* of $\inf_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i)$ of the form $\hat{f}_m(\cdot) = \hat{\varphi}(\cdot, (X_i, Y_i)_{i=1}^m, V)$ for
 - a measurable function $\hat{\varphi} : \mathcal{X} \times (\mathcal{X} \times \mathcal{Y})^m \times \mathcal{V} \rightarrow \mathcal{Y}$ and
 - a $\mathcal{V}$ -valued random element V independent of (X, Y)

data-dependent (and noisy) prediction function $\hat{f}_m$

- *Direct optimization error:* $\frac{1}{m} \sum_{i=1}^m \ell(\hat{f}_m(X_i), Y_i) - \inf_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) \geq 0$

- for classical methods, such as linear regression or logistic regression, one can assume $\hat{f}_m \approx \hat{f}_m^{\mathcal{H}}$
- for modern ML methods, such as boosted trees or neural networks, one typically has $\hat{f}_m \not\approx \hat{f}_m^{\mathcal{H}}$

1.7 Decomposition of the Generalization Error

- **Generalization Error:** $\mathbb{E} \left[\ell \left(\hat{f}_m(X), Y \right) \mid \hat{f}_m \right] = \int_{\mathcal{X} \times \mathcal{Y}} \ell \left(\hat{f}_m(x), y \right) \rho(dx, dy)$
 - $= \mathbb{E} \ell \left(\bar{f}(X), Y \right)$ *irreducible error*
 - $+ \inf_{f \in \mathcal{H}} \mathbb{E} \ell \left(f(X), Y \right) - \mathbb{E} \ell \left(\bar{f}(X), Y \right)$ *approximation error*
 - $+ \mathbb{E} \left[\ell \left(\hat{f}_m^{\mathcal{H}}(X), Y \right) \mid \hat{f}_m^{\mathcal{H}} \right] - \inf_{f \in \mathcal{H}} \mathbb{E} \ell \left(f(X), Y \right)$ *sampling error*
 - $+ \mathbb{E} \left[\ell \left(\hat{f}_m(X), Y \right) \mid \hat{f}_m \right] - \mathbb{E} \left[\ell \left(\hat{f}_m^{\mathcal{H}}(X), Y \right) \mid \hat{f}_m^{\mathcal{H}} \right]$ *indirect optimization error*
(can be positive or negative)
- For fixed $m \in \mathbb{N}$ and increasing $\mathcal{H} \dots$
 - *approximation error* is decreasing (non-increasing)
 - *sampling error* tends to increase (the variance of $\hat{f}_m^{\mathcal{H}}$ is increasing)
 - there is a *tradeoff* between *approximation error* and *sampling error* (similar to bias-variance tradeoff)
 - for *small data*, $\mathcal{H}$ should be simple
 - for *big data*, $\mathcal{H}$ can be complex

1.8 Training Error and Test Error

- Once the *candidate regression function* $\hat{f}_m$ has been chosen, we condition on it

(then it is no longer random)

- training error:* $E_{\text{tr}} = \frac{1}{m} \sum_{i=1}^m \ell(\hat{f}_m(X_i), Y_i)$, *test error:* $E_{\text{te}} = \frac{1}{n} \sum_{i=m+1}^{m+n} \ell(\hat{f}_m(X_i), Y_i)$,

where (X_i, Y_i) , $i = m+1, \dots, m+n$ are independent copies of (X, Y) independent of (X_i, Y_i) ,
 $i = 1, \dots, m$ (*test data*)

- If E_{tr} is significantly smaller than E_{te} , the data most likely was *overfitted*

- Sample variance* $\sigma_{\text{te}}^2 = \frac{1}{n-1} \sum_{i=m+1}^{m+n} \left(\ell(\hat{f}_m(X_i), Y_i) - E_{\text{te}} \right)^2$

- If $\mathbb{E} \ell^2(\hat{f}_m(X), Y) < \infty$, it follows from the *Law of Large Numbers* that $\sigma_{\text{te}}^2 \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \text{Var}(\ell(\hat{f}_m(X), Y))$

- By the *Central Limit Theorem*: $E_{\text{te}} - \mathbb{E} \ell(\hat{f}_m(X), Y) \approx \mathcal{N}\left(0, \frac{\sigma_{\text{te}}^2}{n}\right)$

- Therefore, $\mathbb{P}\left[E_{\text{te}} - \frac{\sigma_{\text{te}}}{\sqrt{n}} q_{1-\alpha/2} \leq \mathbb{E} \ell(\hat{f}_m(X), Y) \leq E_{\text{te}} + \frac{\sigma_{\text{te}}}{\sqrt{n}} q_{1-\alpha/2}\right] \approx 1 - \alpha$ for $\alpha \in (0, 1)$,

where $q_{1-\alpha/2}$ is the standard normal $(1 - \alpha/2)$ -quantile

1.9 Regularization

- If there is *overfitting*; e.g. $E_{\text{tr}} < 0.85 \times E_{\text{te}}$, one can reduce the *complexity of the hypothesis class* $\mathcal{H}$
- Alternatively, one can *regularize* the empirical loss minimization problem by *penalizing* the complexity of potential prediction functions $f \in \mathcal{H}$:

$$\inf_{f \in \mathcal{H}} \left(\frac{1}{m} \sum_{i=1}^m \ell(f(X_i), Y_i) + \lambda_1 \|f\|_1 + \lambda_2 \|f\|_2^2 \right) \quad \text{for hyperparameters } \lambda_1, \lambda_2 \geq 0 \quad (*)$$


LASSO ridge regression
└──┘
ElasticNet

- Let $\hat{f}_m^{(\lambda_1, \lambda_2)}$ be a numerical solution of $(*)$ and $E_{\text{te}}^{(\lambda_1, \lambda_2)} = \frac{1}{n} \sum_{i=m+1}^{m+n} \ell \left(\hat{f}_m^{(\lambda_1, \lambda_2)}(X_i), Y_i \right)$
- Find $\lambda_1, \lambda_2 \geq 0$ such that $E_{\text{te}}^{(\lambda_1, \lambda_2)}$ is minimal (if $E_{\text{te}}^{(\lambda_1, \lambda_2)} < E_{\text{te}}$, regularization helps)
- There exist different regularization methods, e.g. *early stopping (see later)*